CORRIGENDUM TO "ON GRADED RINGS WITH HOMOGENEOUS DERIVATIONS"

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Abstract. We correct several points in [1]. We give a corrected proof of Proposition 2.7 under an additional hypothesis and record its consequences for the results that depend on it. We also replace Examples 3.4 and 3.6(2), clarify the proof of Theorem 4.1, correct a reference in the proof of Theorem 4.3, and correct Theorem 4.4.

Numbering refers to the original paper [1]. As in [1], the grading group G is assumed to be abelian.

1. Proposition 2.7

In the proof of Proposition 2.7, the relation $[d(r), r] = 0$ is established for homogeneous $r \in I$, and the same conclusion is then asserted for arbitrary $x \in I$. This requires an additional argument, since for $x = \sum_{g\in F} x_g$, the expression $[d(x), x]$ also contains the mixed terms $[d(x_g), x_h] + [d(x_h), x_g]$, with $g \neq h$.

The proposition remains valid under the additional hypothesis

$$R \text{ is 2-torsion free, or } G \text{ has no element of order 2.} \tag{\textbf{H}}$$

Lemma 1.1. *Let R be a gr-semiprime ring satisfying* (**H**)*. If $z = b_1 + b_2 \in Z(R)$, where b_1, b_2 are homogeneous and $z^2 = 0$, then $z = 0$.*

Proof. If $\deg b_1 = \deg b_2$, then z is homogeneous, and the result follows from Proposition 2.2(1).

Assume $\deg b_1 = \alpha \neq \gamma = \deg b_2$. By [3, Lemma 2.2], $Z(R)$ is graded. Hence $b_1, b_2 \in Z(R)$ and

$$b_1^2 + 2b_1b_2 + b_2^2 = 0. \tag{1}$$

If R is 2-torsion free, the component of degree $\alpha\gamma$ gives $b_1b_2 = 0$. If $\alpha^2 \neq \gamma^2$, then $b_1^2 = b_2^2 = 0$; if $\alpha^2 = \gamma^2$, then $b_1^3 = 0$. In either case $b_1 = b_2 = 0$ by Proposition 2.2(1).

If G has no element of order 2, then $\alpha^2, \alpha\gamma, \gamma^2$ are pairwise distinct, so $b_1^2 = b_2^2 = 0$, and again $b_1 = b_2 = 0$. □

Corrected proof of Proposition 2.7. Set $f(x) = [d(x), x]$. Since $f(x) \in Z(R)$, linearization gives

$$[d(x), y] + [d(y), x] \in Z(R) \qquad (x, y \in I). \tag{2}$$

Taking $y = x^2$ gives $4f(x)^2 = 0$. Taking $y = d(x)x$ gives

$$3f(x)d(x) + d(f(x))x \in Z(R),$$

and hence $3f(x)^2 = 0$. Therefore $f(x)^2 = 0$ for every $x \in I$.

If $r \in I \cap \mathcal{H}(R)$, then $f(r)$ is homogeneous, central and nilpotent, so $f(r) = 0$ by Proposition 2.2(1). If $r, s \in I \cap \mathcal{H}(R)$, then $f(r + s) = [d(r), s] + [d(s), r]$ and $f(r + s)^2 = 0$; hence Lemma 1.1 gives $f(r + s) = 0$.

Finally, for $x = \sum_{g\in F} x_g$,

$$f(x) = \sum_{g\in F} f(x_g) + \sum_{\substack{\{g,h\}\subseteq F \\ g\neq h}} \big([d(x_g), x_h] + [d(x_h), x_g]\big),$$

and every term vanishes. Thus $[d(x), x] = 0$ for all $x \in I$.

Therefore Proposition 2.7 holds under (**H**). Consequently, Theorems 3.3 and 4.1 and Corollary 4.2 remain valid under (**H**). If d is already commuting on I, no additional hypothesis is required.

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2. Examples 3.4 and 3.6(2)

The decomposition used in Example 3.4 does not define a grading. For $A = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$, one has $(X, A) \in R_1$, whereas $(X, A)^2 = (X^2, I_2) \notin R_2$.

Moreover, the ring $\left\{\left(\begin{smallmatrix} a & b \\ b & a \end{smallmatrix}\right) : a, b \in \mathbb{F}\right\}$ used in Example 3.6(2) is commutative.

The examples should be replaced as follows.

Example 2.1 (Example 3.4, corrected)**.** Let $R = \mathbb{C}[X] \times M_2(\mathbb{C})$ with the $\mathbb{Z}$-grading

$$\begin{aligned} R_0 &= \mathbb{C} \times \left\{ \begin{pmatrix} z & 0 \\ 0 & z' \end{pmatrix} : z, z' \in \mathbb{C} \right\}, \\ R_1 &= \operatorname{span}_{\mathbb{C}}(X) \times \mathbb{C}e_{12}, \qquad R_{-1} = \{0\} \times \mathbb{C}e_{21}, \\ R_n &= \operatorname{span}_{\mathbb{C}}(X^n) \times \{0\}, \qquad n \geq 2, \\ R_n &= \{0\}, \qquad n \leq -2. \end{aligned}$$

Define $d(P, M) = (P', 0)$. Then d is a nonzero homogeneous derivation with $d(R_n) \subseteq R_{n-1}$, and $[d(x), x] = 0$ for every $x \in R$. The ring R is not gr-prime, since $(1, 0)R(0, I_2) = 0$, while it is noncommutative.

Example 2.2 (Example 3.6(2), corrected)**.** Let $\mathbb{F}$ be a field of characteristic 2, and let $R = M_2(\mathbb{F})$ with the $\mathbb{Z}_4$-grading

$$R_0 = \left\{ \begin{pmatrix} a & 0 \\ 0 & e \end{pmatrix} : a, e \in \mathbb{F} \right\}, \qquad R_1 = \mathbb{F}e_{21}, \qquad R_2 = 0, \qquad R_3 = \mathbb{F}e_{12}.$$

Let $d = \operatorname{ad}_{e_{12}}$. Then d is homogeneous and nonzero, and for $x = \left(\begin{smallmatrix} a & b \\ c & e \end{smallmatrix}\right)$,

$$d(x) = cI_2 + (a + e)e_{12}.$$

Hence $[d(x), d(y)] = 0$ for all $x, y \in R$.

Since R is prime and noncommutative, taking $d_1 = d_2 = d$ shows that the hypothesis $\operatorname{char} R \neq 2$ in Theorem 3.5 cannot be omitted.

3. Theorem 4.1

Under (**H**), Proposition 2.7 gives $[d(x), x] = 0$ for all $x \in I$. The proof then proceeds as in [2, Theorem 3], with the following clarification.

The gr-prime argument is applied separately to each gr-prime ideal P_i. A gr-prime ideal is called type $(*)$ if $[I, I] \subseteq P_i$, and type $(**)$ otherwise. In the latter case, the argument gives $Id(I) \subseteq P_i$ and then $d(I) \subseteq P_i$.

Let P_{Γ_1} and P_{Γ_2} denote the intersections of the ideals of type $(*)$ and type $(**)$, respectively, and let J be the graded left ideal generated by $d(R)I$. Then

$$[J, J] \subseteq P_{\Gamma_1}, \qquad J \subseteq P_{\Gamma_2}.$$

Since $P_{\Gamma_1} \cap P_{\Gamma_2} = 0$, we obtain $[J, J] = 0$. The remainder of the original proof shows that $J \neq 0$ and $J \subseteq Z(R)$.

Thus Theorem 4.1 remains valid under (**H**). If d is already commuting on I, no additional hypothesis is required.

4. Theorem 4.3

In the proof of Theorem 4.3, the reference to Proposition 2.7 in $I = Z(I) \subseteq Z(R)$ should be replaced by Proposition 2.2(3).

5. Theorem 4.4

Theorem 4.4 is false without a 2-torsion-freeness assumption. Indeed, let R and d be as in Example 2.2, and take $I = R$. Then R is prime, hence gr-semiprime, $d \neq 0$, and $[d(x), d(y)] = 0$ for all $x, y \in R$. However, $R = M_2(\mathbb{F})$ is simple and noncommutative, and therefore contains no nonzero central ideal.

The corrected statement is as follows.

Theorem 5.1 (Corrected Theorem 4.4)**.** *Let R be a 2-torsion-free gr-semiprime ring and I a nonzero graded ideal of R. Assume that R admits a homogeneous derivation d which is nonzero on I and satisfies $[d(x), d(y)] = 0$ for all $x, y \in I$. Then R contains a nonzero central graded ideal.*

Proof. As in the original proof, one obtains

$$[d(x), y]zd^2(t) = 0 \qquad (x, y, z, t \in I). \tag{3}$$

Let $\{P_i : i \in \Lambda\}$ be a family of gr-prime ideals with zero intersection. Arguing modulo each P_i, using Proposition 2.1(4) on homogeneous elements, gives

$$[d(I), I] \subseteq P_i \qquad \text{or} \qquad d^2(I) \subseteq P_i.$$

In the second case, $d^2(xy) = d^2(x)y + 2d(x)d(y) + xd^2(y)$ and the same gr-prime argument give $2d(I) \subseteq P_i$. Thus, in either case, $2[d(x), y] \in P_i$ for all $x, y \in I$.

Hence $2[d(x), y] = 0$ for all $x, y \in I$. Since R is 2-torsion free, $[d(x), y] = 0$ for all $x, y \in I$. Therefore d is commuting on I, and the commuting case of Theorem 4.1 yields a nonzero central graded ideal of R. □

References

[1] Y. Ait Mohamed, *On graded rings with homogeneous derivations*, Rev. Un. Mat. Argentina **69** (2026), no. 1, 55–69. https://doi.org/10.33044/revuma.4934

[2] H. E. Bell and W. S. Martindale III, *Centralizing mappings of semiprime rings*, Canad. Math. Bull. **30** (1987), no. 1, 92–101. https://doi.org/10.4153/CMB-1987-014-x

[3] J. Öinert, *Central idempotents in group-graded rings*, arXiv:2605.20008 [math.RA] (2026). https://doi.org/10.48550/arXiv.2605.20008

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